

Math 271 Summary So Far

So far we've covered elements of discussion and creating proof statements for math. We started with understanding some of the symbols that would be seen in math, and how sets are used in these proofs to create the assumed statements we are either stating or proving. Some symbols like $x \in A$ mean that x is a subset of A , whether it's cardinality ($|x|$) is 1 or more. $X \subset A$ would mean the set X is a subset of A and $X \subseteq A$ would mean that X is still a subset of A that might also have the same cardinality of A .

After understanding the sets we then talked about how to prove statements. So far we've covered three: Direct Proof, Proof by Contrapositive, and Proof by Contradiction. Direct proof is when you mathematically show that the statement is true. This can be with an example value or variables that would suffice for general statements. Proof by Contrapositive shows that if Statement A then Statement B , then if the opposite of Statement B then the opposite of Statement A . This works when the contrapositive is easier to mathematically prove. Proof by Contradiction is by proving that the contradiction is false. This, again, is something to approach if the statement itself is difficult to show proof mathematically. By showing the contradiction is false that means the original statement would have to be true.

So far this is what we've gone over. It's very valuable with dealing with sets of data and deriving truth based on what cases or patterns can be made.